

Internship Problem -

Siemens.Internship2026.GradeBook

I. SOLID principle violation

1. Which SOLID principle is violated: SRP – Single Responsibility Principle

Where in the code (file + line/method): ItemController.cs - methods GetAll()

Why it is a violation: The definition of SRP says: “A class should have one and only one reason to change, meaning that a class should have only one job.”. In our case, the controller had two jobs: handling HTTP request and also, filtering grades and computing statistics.

The fix you applied: I moved the filter function and computing statistics into a Service Layer (ItemService). Now, the controller has a single responsibility: to receive the HTTP request and return the response.

2. Which SOLID principle is violated: DIP – Dependency Inversion Principle

Where in the code (file + line/method): ItemController.cs - methods GetAll() and GetById(), where Console.WriteLine() was called directly

Why it is a violation: The definition of DIP says: “High-level modules should not depend on low-level modules. Both should depend on abstractions.” In the initial code, the controller was using Console.WriteLine(). This way, the controller was strictly tied to the system console.

The fix you applied: I used the ILogger<ItemController> interface and replaced Console.WriteLine() with _logger.LogInformation/LogWarning . Now, the controller depends on an abstraction (the ILogger interface).

II. Upgrade the project from .NET 8 to .NET 10.

I changed in ‘Siemens.Internship2026.GradeBook.csproj’

From: <TargetFramework>net8.0</TargetFramework>

To: <TargetFramework>net10.0</TargetFramework>

III. **Service Layer**

I created a function named `GetFirstNGrades()` that retrieves only the first N grades that meet both of the following criteria: - The grade is a passing grade (greater than or equal to 5) where the grade is active N is a parameter provided by the user.

IV. **Repository Refactoring**

The link that I was provided with didn't work and I couldn't access it. In order to test the program, I created my own JSON list using Pastebin.